KOST-VAL

Manual

Content

1 Preface

KOST-Val is a Java-based application for validating the structure and content of PDF/A, JP2, JPEG, TIFF, PNG, FLAC, WAVE, MP3, MKV, MP4, XML, SIARD files as well as Submission Information Packages (SIP) for digital information ingest. It is an open source application under a GPL v3+ licence. KOST-Val uses unmodified components of other manufacturers by embedding them directly into the source code. Users of KOST-Val are requested to adhere to these components ‘terms of licence. Please refer to chapter 9 for further information.

The results (including information on inconsistencies and errors) are output for every step and written into a validation log.

The validation steps are executed sequentially. Whenever possible the validation shall continue after an error has been detected in order to reduce the number of correction cycles.

From version 2.3.0.3 KOST-Val has additional repair functions built in.
If these repair functions are used, the person using them is responsible for quality control, revalidation and any further use, including documentation.

2 System requirements

- 64bit Microsoft Windows
- At least 512 MB RAM
- At least 20 GB hard disk space

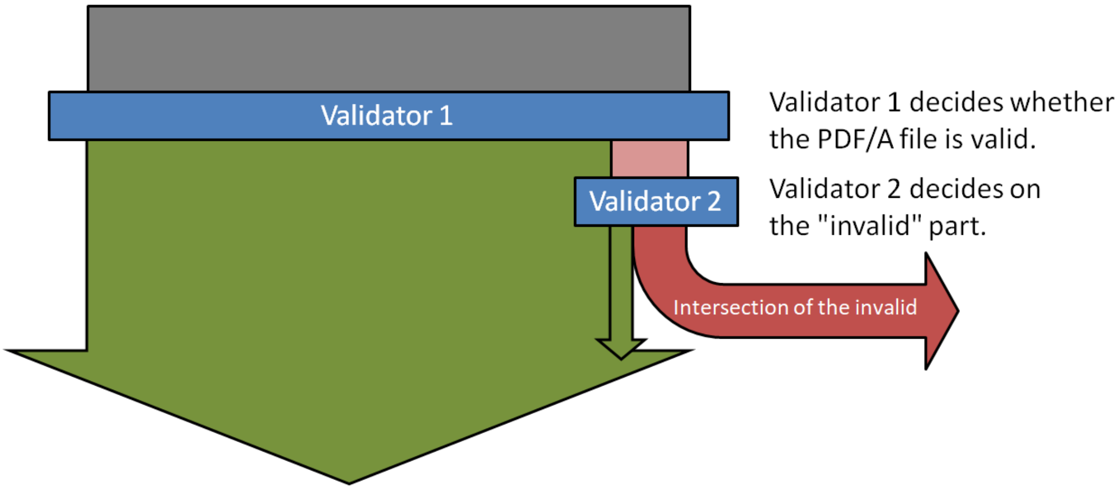
3 Open issues / Feedback

Open issues ranging including bugs, requested features, and questions, are listed on the software development platform GitHub at https://github.com/KOST-CECO/KOST-Val/issues and can also be communicated to kost-val@kost-ceco.ch.

These issues are managed by the development team. Any and all contributions are welcome.

4 Introduction dual PDF/A validation

For PDF/A KOST-Val offers the possibility of a dual validation. A PDF/A file is first checked by a first validator. If the result is invalid, a check by a second validator follows. The PDF/A file is considered valid if at least one of the validators identifies it as valid, and as invalid if both validators identify it as invalid.[1]



Dual PDF/A validation may only be used if the archive allows potentially invalid PDF/A files to be accepted. If this is not the case, dual PDF/A validation should be avoided.

Both 3-Heights™ PDF/A Validator from PDF-Tools and veraPDF are used for dual validation. If only one validator is activated, a single validation is automatically performed.

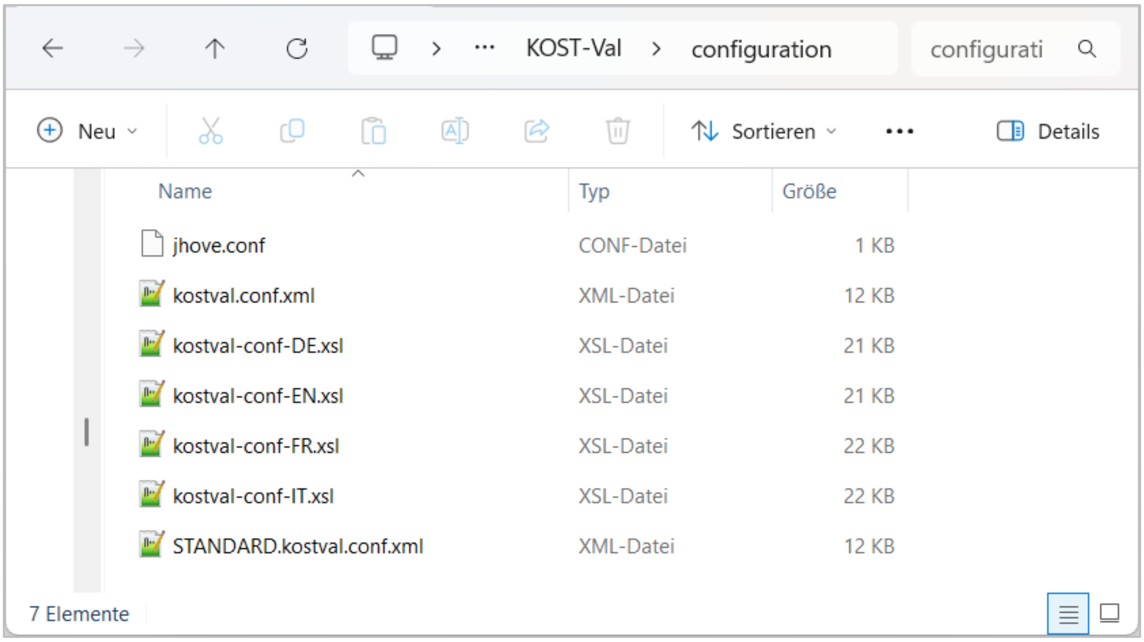
The conceptual basis for dual validation is the observation that even high-quality PDF/A validators can produce different results. This is due on the one hand to the fact that the actual PDF/A standard includes a set of other standards which are not necessarily implemented in the validators in every detail. On the other hand, certain specifications of the standard are formulated in such a way that they can legitimately be implemented in various ways. The fact that all relevant tools implement the specification uniformly and completely remains a pipe dream for the time being. Therefore KOST-Val offers dual validation as an interim solution.

[1] The dual validation can only be carried out with high-quality PDF/A validators in this sense. These high requirements are fulfilled among others by the latest versions of 3-Heights™ PDF/A Validator from PDF-Tools and veraPDF.

5 Installation of KOST-Val

1	KOST-Val (version 2.1.3.0 and newer) is only offered in the 64bit installation package KOST-Tools.msi[2]. https://github.com/KOST-CECO/KOST-Val/releases/latest After downloading KOST-Tools the installation package must be executed with administration rights. KOST-Val is then available in the start menu under KOST-Tools.

6 Configuration of KOST-Val



The "configuration" folder contains the file "jhove.conf" that do not need adjustment.

"jhove.conf" is used for the internal validation by JHOVE.

The configuration file "kostval.conf.xml", “STANDARD.kostval.conf.xml” as well as the four style sheets are copied to the directory "USERHOME/.kost-val_2x/configuration" if not correct or currently available. All configurations of the KOST-Val can be made via GUI.

[2] More detailed instructions on the installation and its scope can be found in the KOST-Tools manual.

6.1 Parts of the configuration file "kostval.conf.xml"

The configuration file "kostval.conf.xml" consists of several parts.
The following is a short description of the configuration parts.

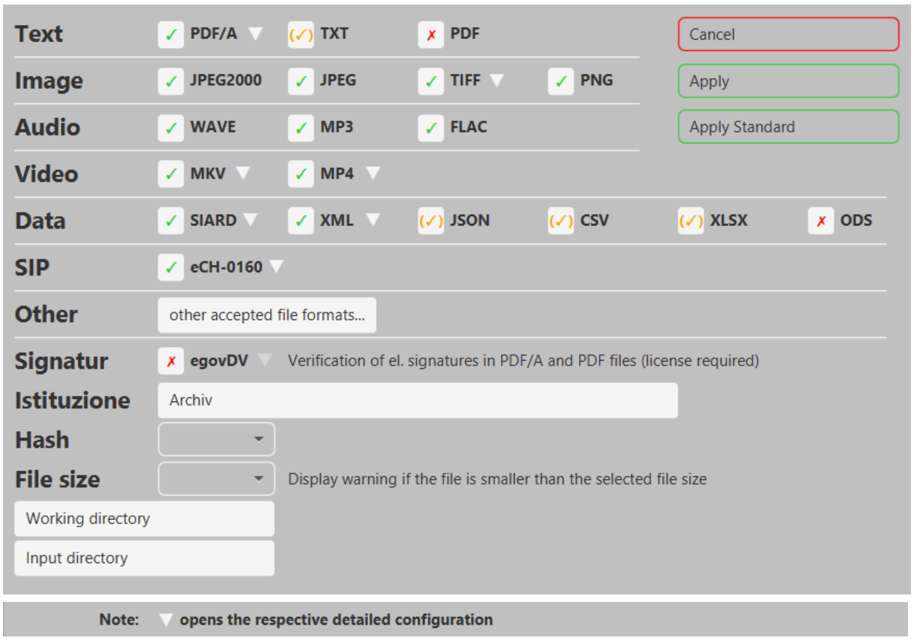
KOST-Val **Configuration**

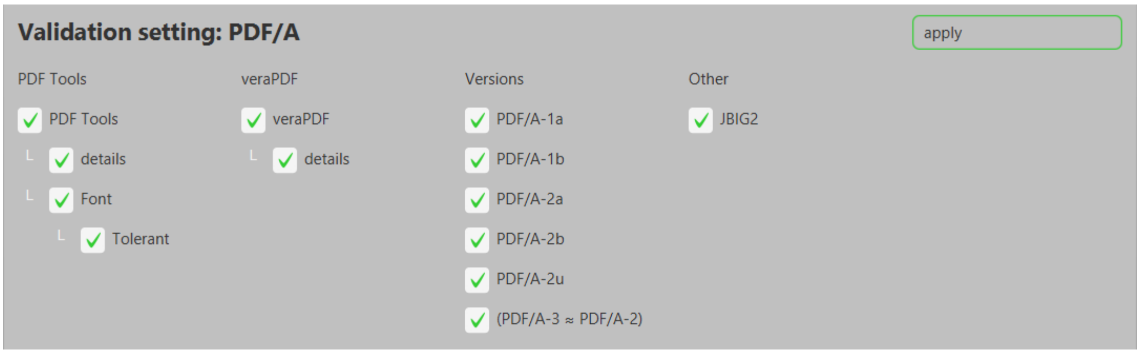
Legend:

✓ = accepted and validate
(✓) = accepted
✗ = not accepted

PDF/A: Acceptance and validation [✓]	✓
...	
TXT: Acceptance [(✓)]	(✓)
PDF: Acceptance [✗]	✗
JPEG2000: Acceptance and validation [✓]	✓
JPEG: Acceptance and validation [✓]	✓
TIFF: Acceptance and validation [✓]	✓
...	
PNG: Acceptance and validation [✓]	✓
FLAC: Acceptance [✓]	✓
WAVE: Acceptance [✓]	✓
MP3: Acceptance [✓]	✓
MKV: Acceptance and validation [✓]	✓
...	
MP4: Acceptance and validation [✓]	✓
...	
XML: Acceptance and validation [✓]	✓
...	
JSON: Acceptance [(✓)]	(✓)
SIARD: Acceptance and validation [✓]	✓
...	
CSV: Acceptance [(✓)]	(✓)
XLSX: Acceptance [(✓)]	(✓)
ODS: Acceptance [✗]	✗
SIP: Validation [✓]:	✓
...	
Other accepted file formats [WARC, HTML, DWG]:	HTML WARC DWG
Signatur in PDF/A- und PDF-Dateien: Prüfung [✗]	✗
...	
Name of the institution [Archiv]:	Archiv
Calculate and output hash value of files. Empty means no calculation and output []:	
Issue warning for small files. Empty means no warning []:	
Working directory []:	
Input directory []:	

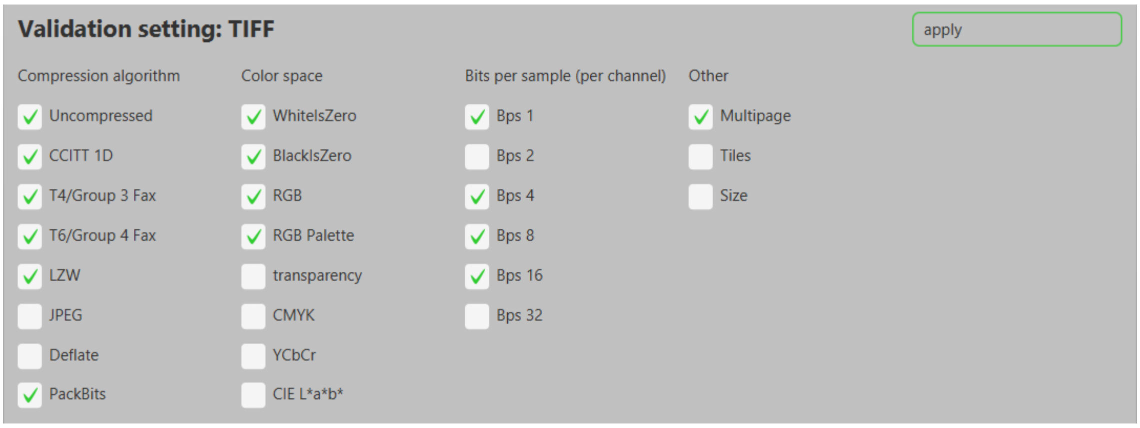
!	The user of the repair function must assume responsibility for quality assurance, revalidation, and any further use, including documentation!



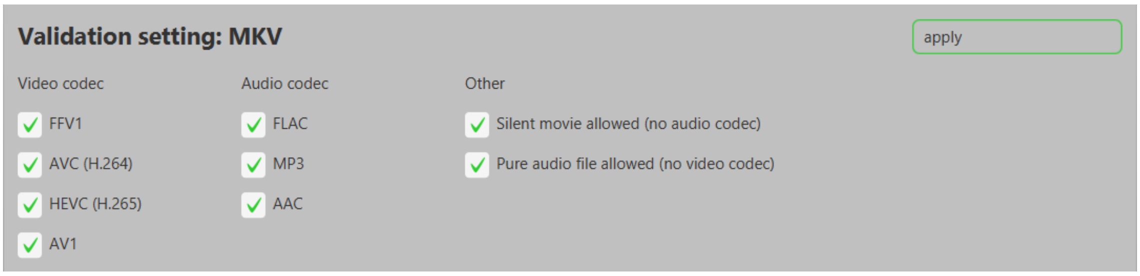


PDF/A: Acceptance and validation [✓]	✓
PDF/A validation with PDF Tools [yes]:	yes
- PDF Tools also detailed errors in English [yes]:	yes
- Validation (searchability and extractability) with PDF Tools [tolerant]:	tolerant
PDF/A validation with veraPDF [yes]:	yes
- veraPDF also detailed errors in English [yes]:	yes
Allowed PDF/A versions [1A, 1B, 2A, 2B, 2U]:	1A 1B 2A 2B 2U
Validate PDF/A-3 to PDF/A-2 and generate warning message instead of error [yes]:	yes
JBIG2 compression allowed [yes]:	yes
Create a repaired PDF/A copy [no]:	no
- repair invalid PDF/A and unacceptable PDF files to PDF/A-2u [no]:	no

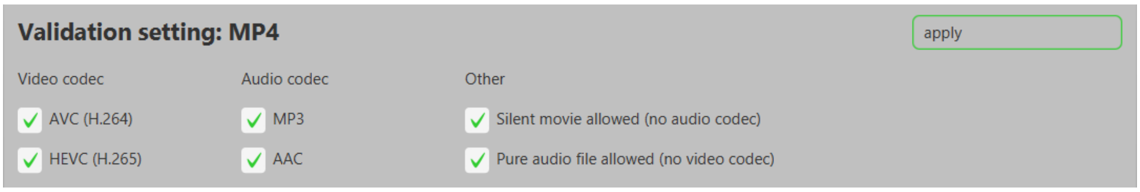
The repair function for rejected PDF and invalid PDF/A files requires the installation of "dmstools PDF/A Converter" v1.7.0.1 and a valid license.
The internal license is available only to member institutions and supporters. The annual KOST license code must be requested from the office and stored under "USERHOME/.kost-val_2x/configuration".



TIFF: Acceptance and validation [✓]	✓
Allowed compression algorithms [uncompressed, CCITT 1D, CCITT Group 3, CCITT Group 4, LZW, PackBits]:	uncompressed CCITT 1D CCITT Group 3 CCITT Group 4 LZW PackBits
Allowed color spaces [white is zero, black is zero, RGB, palette color]:	white is zero black is zero RGB palette color
Bits per sample allowed [1, 4, 8, 16]:	1 4 8 16
Multipage TIFFs allowed [yes]:	yes
Build in tiles allowed [no]:	no
File sizes of 1000MB (~1GB) and larger allowed [no]:	no

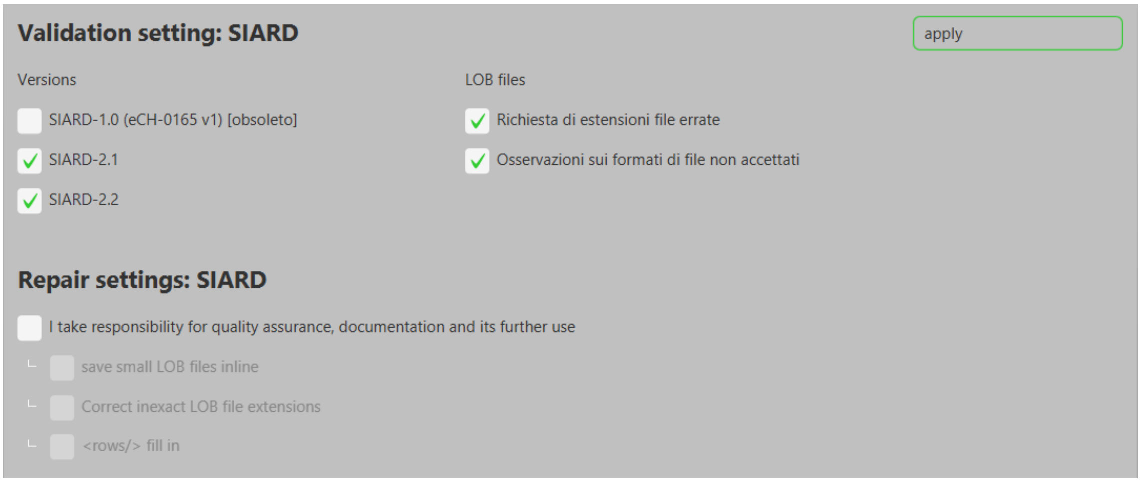


MKV: Acceptance and validation [✓]	✓
- Allowed video codec [FFV1, AVC, HEVC, AV1]:	FFV1 AVC HEVC AV1
- Allowed audio codec [FLAC, MP3, AAC]:	FLAC MP3 AAC
- Silent movie allowed (no audio codec) [Warning]:	Warning
- Pure audio file allowed (no video codec) [Warning]:	Warning



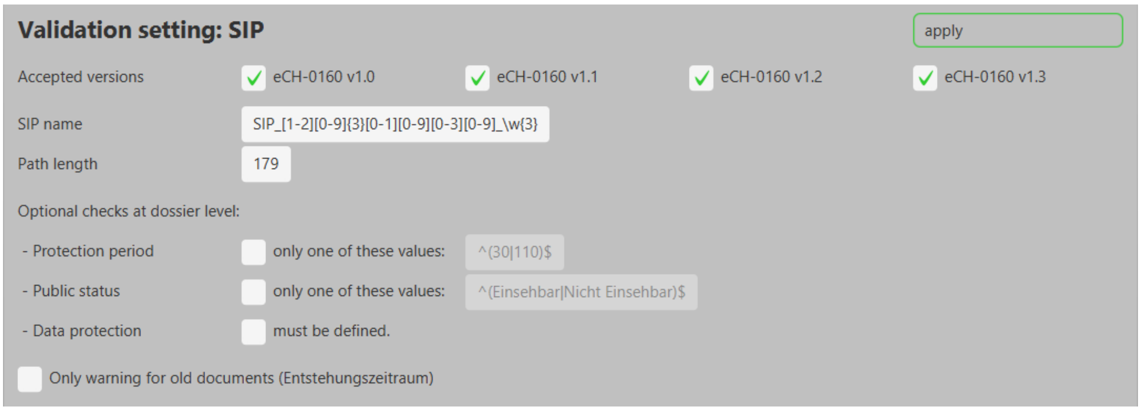
MP4: Acceptance and validation [✓]	✓
- Allowed video codec [AVC, HEVC]:	AVC HEVC
- Allowed audio codec [MP3, AAC]:	MP3 AAC
- Silent movie allowed (no audio codec) [Warning]:	Warning
- Pure audio file allowed (no video codec) [Warning]:	Warning

XML: Acceptance and validation [✓]	✓
Schema validation [no]:	no

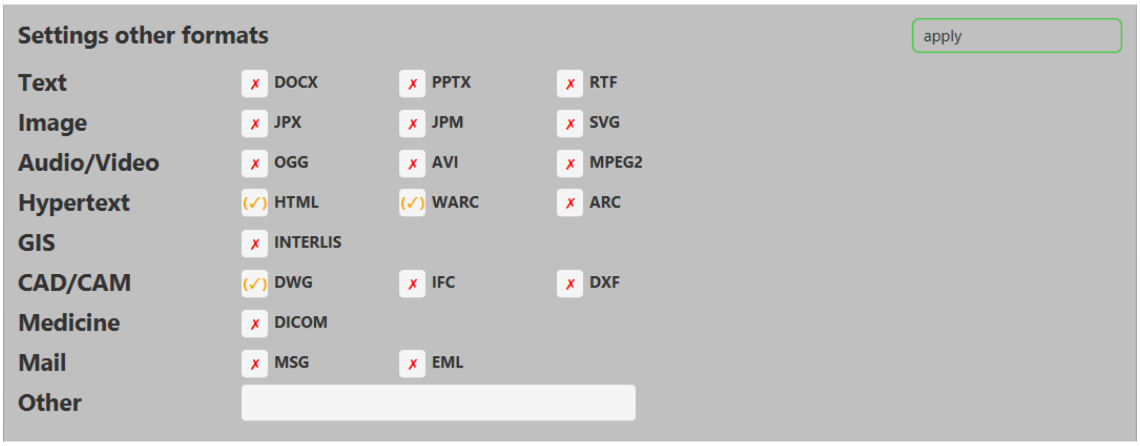


SIARD: Acceptance and validation [✓]	✓
Allowed SIARD versions [2.1, 2.2]:	2.1 2.2
Claim for incorrect file extensions [Error]:	Error
Remarks on non-accepted file formats [Error]:	Error
Validation of accepted file formats [Validate]:	
Create a repaired SIARD copy [no]:	no
- save small LOB files inline [no]:	no
- correct inexact LOB file extensions [no]:	no
- fill in number of rows in metadata.xml [no]:	no

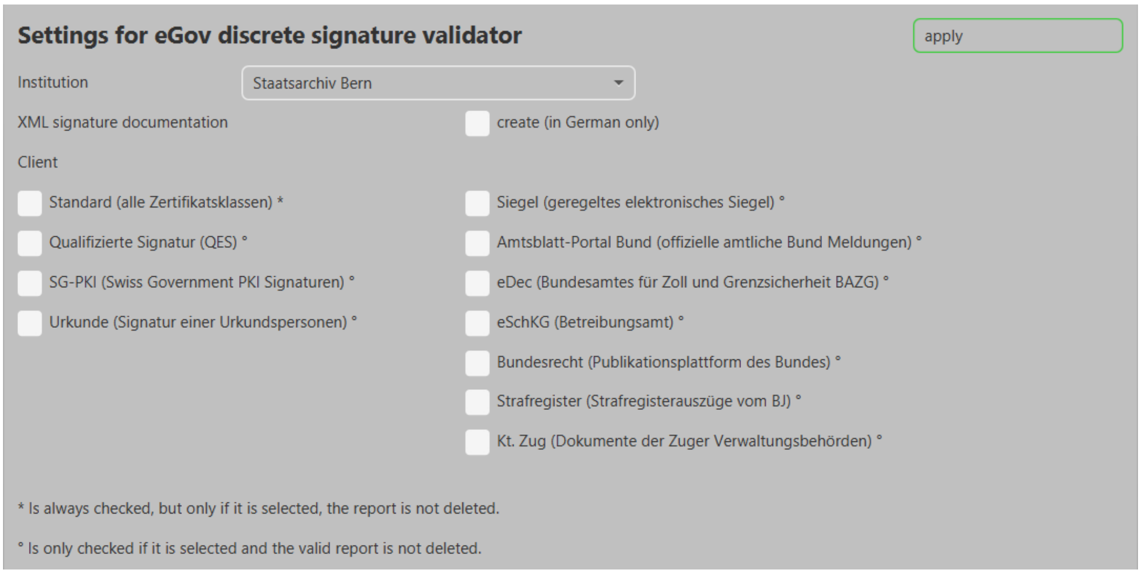
If the repair of SIARD files is activated and a repair is carried out, the new SIARD file is written to the output folder "USERHOME/.kost-val_2x/ OUTPUT".
Corresponding information is also written to the log.
The original file is deliberately not overwritten.



SIP: Validation [✓]:	✓
Allowed eCH-0160 SIP versions [1.0, 1.1, 1.2, 1.3]:	1.0 1.1 1.2 1.3
Allowed maximum number of characters in path lengths [179]:	179
Specifications for the structure of the SIP name [SIP_[1-2][0-9]{3}[0-1][0-9][0-3][0-9]_\w{3}]:	SIP_[1-2][0-9]{3}[0-1][0-9][0-3][0-9]_\w{3}
Optional checks at dossier level:	
- Check regarding protection period [no] [^(30|110)$]:	no ^(30|110)$
- Check regarding public status [no] [^(Einsehbar|Nicht Einsehbar)$]:	no ^(Einsehbar|Nicht Einsehbar)$
- Check data protection [no]:	no
Only warning for old documents (Entstehungszeitraum) [no]:	no



Other accepted file formats [WARC, HTML, DWG]:	HTML WARC DWG



Signatur in PDF/A- und PDF-Dateien: Prüfung [✗]	✗
Institution, welche die Prüfung durchführt []:	
XML-Signaturdokumentation erstellen [no]:	no
Report, welcher gesichert wird, wenn yes:	
- Standard (alle Zertifikatsklassen) [no]:	no
Mandante, welche geprüft und gesichert werden, wenn valide und yes:	
- Qualifizierte Signatur (QES) [no]:	no
- SG-PKI (Swiss Government PKI Signaturen) [no]:	no
- Urkunde (Signatur einer Urkundspersonen) [no]:	no
- Siegel (geregeltes elektronisches Siegel) [no]:	no
- Amtsblatt-Portal Bund (offizielle amtliche Meldungen) [no]:	no
- eDec (Bundesamtes für Zoll und Grenzsicherheit BAZG) [no]:	no
- eSchKG (Betreibungsamt) [no]:	no
- Bundesrecht (Publikationsplattform des Bundes) [no]:	no
- Strafregister (Strafregisterauszüge vom BJ) [no]:	no
- Kt. Zug (Dokumente von den Zuger Verwaltungsbehörden) [no]:	no

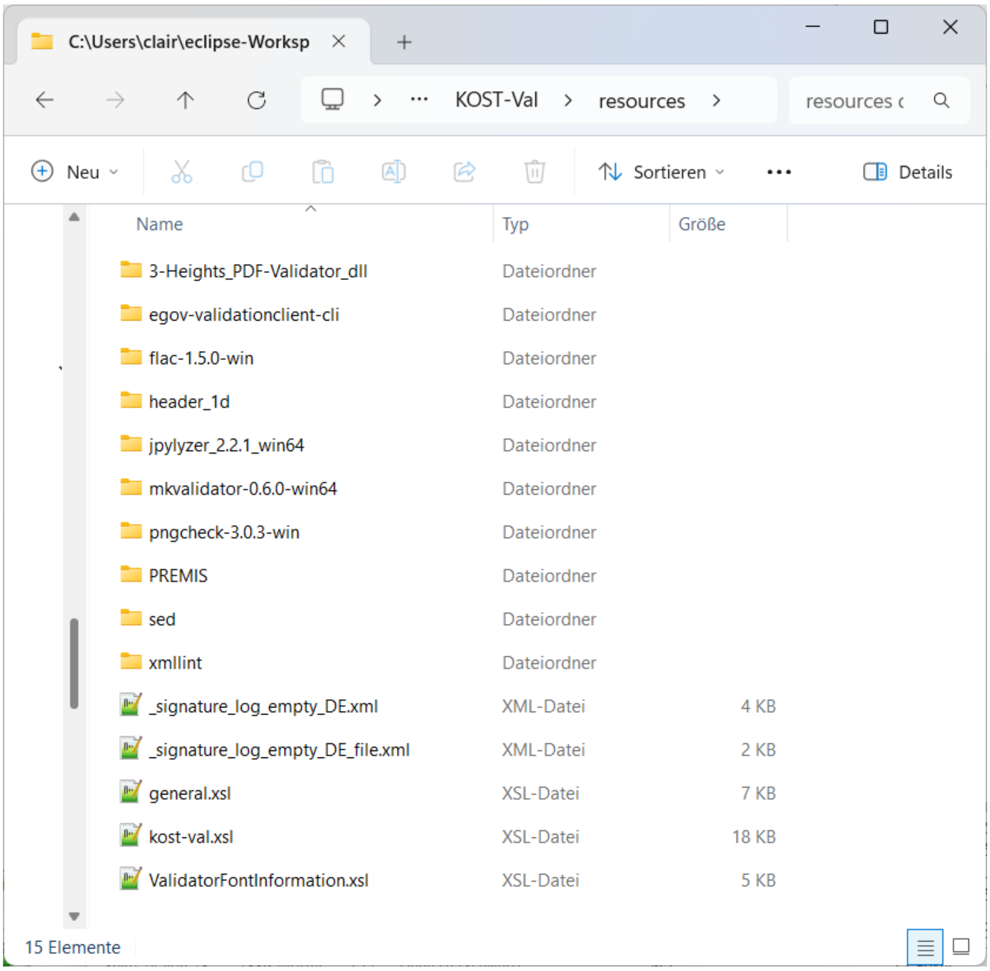
The discrete validator only works with a cantonal license, which must be integrated into KOST-Val (see chapter 9.2 egov-validationclient-cli license).

This function is currently only available in German and for the following institutions:

AG:	Staatsarchiv Aargau		
BS:	Staatsarchiv Basel-Stadt		
BE:	Staatsarchiv Bern	Stadtarchiv Bern	Burgerbibliothek Bern
GR:	Staatsarchiv Graubünden		
LU:	Staatsarchiv Luzern	Stadtarchiv Luzern	
SG:	Staatsarchiv St. Gallen	Stadtarchiv St. Gallen	
TG:	Staatsarchiv Thurgau		

7 Resources of KOST-Val

All resources of KOST-Val are stored in the subfolder "resources".

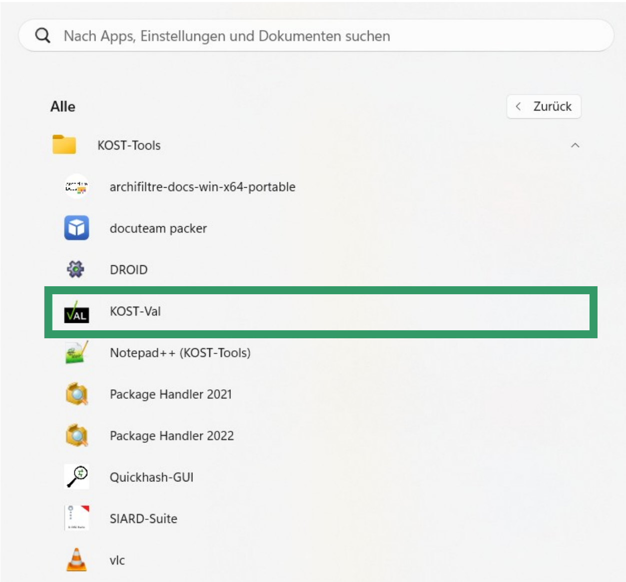


8 Start the validation

!	KOST-Val is not thread safe! That is to say that concurrent instances of KOST-Val cannot be executed without interfering with each other. Concurrent execution of KOST-Val may lead to errors such as a missing working copy.

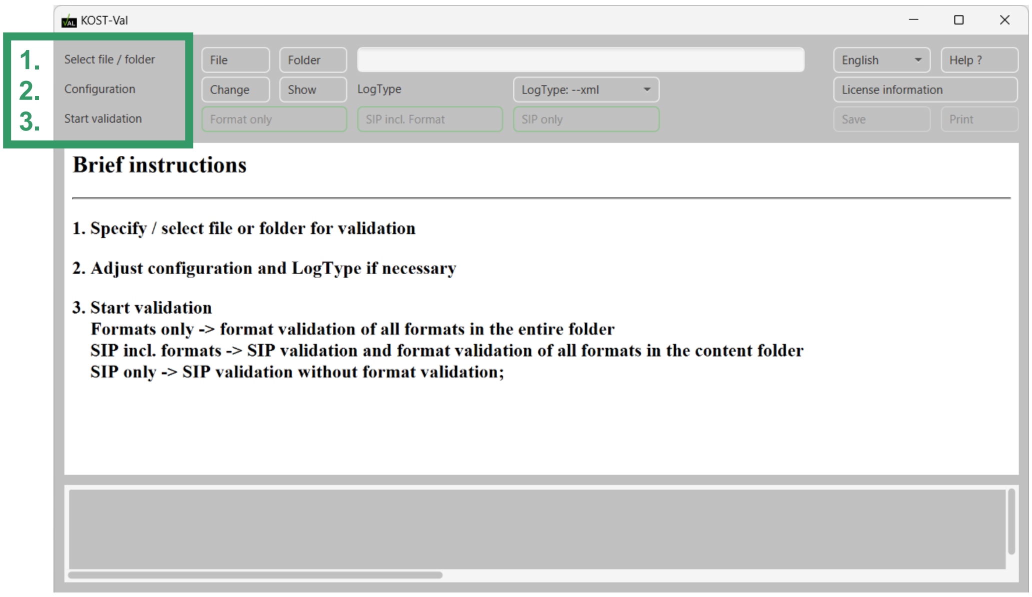
8.1 KOST-Val GUI

1 Start KOST-Val by clicking on “KOST-Val” in the “KOST-Tools” start menu.

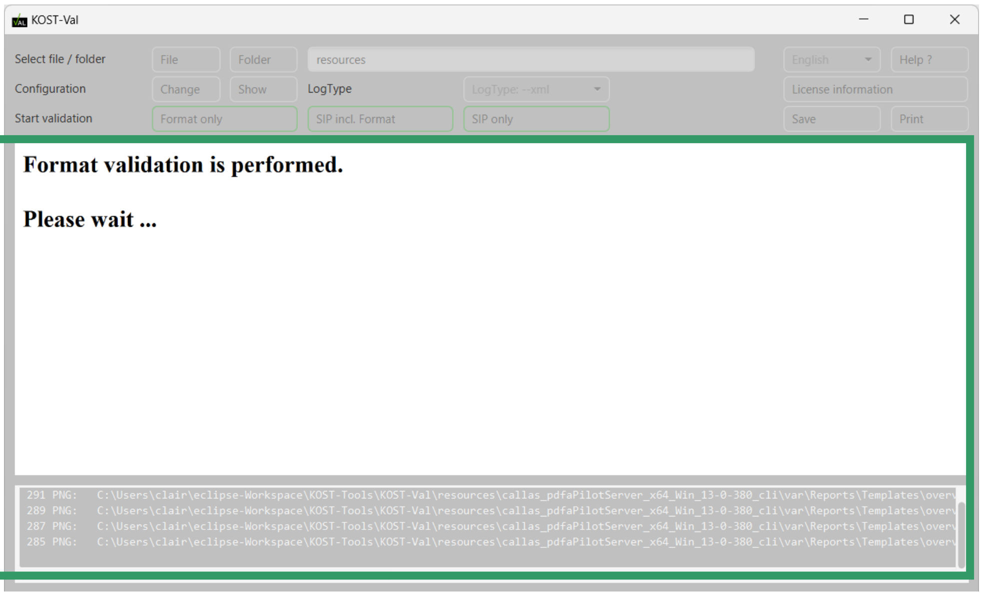


2

1. Specify / select file or folder for validation
2. Adjust configuration and LogType if necessary
3. Start validation

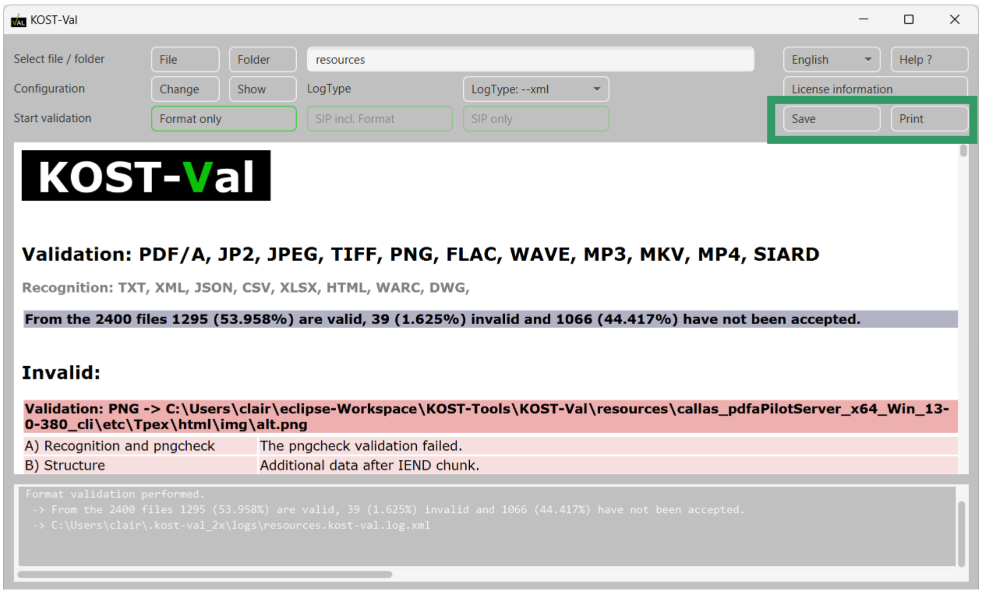


4. Wait until the validation is finished.
The progress / current file is shown in the field below.



3

At the end of the validation the log file is displayed. This log file contains additional detailed information about the individual invalid validation steps, in particular the affected validation step and the corresponding error. If desired, the KOST-Val-Log file can be saved or printed.



8.2 Start the validation manually

1

Open a command prompt and change to the desired working directory
(`cd "C:\Program Files\KOST-CECO\KOST-Tools\KOST-Val"`)[3].

```
Eingabeaufforderung

C:\Users>cd "C:\Program Files\KOST-CECO\KOST-Tools\KOST-Val"

C:\Program Files\KOST-CECO\KOST-Tools\KOST-Val>
```

2

Invoke the KOST-Val command (separate command options with spaces).
`..\Liberica_JRE\bin\java.exe -jar cmd_KOST-Val.jar --format resources --en --xml`

```
C:\Program Files\KOST-CECO\KOST-Tools\KOST-Val>..\Liberica_JRE\bin\java.exe -jar cmd_KOST-Val.jar --format resources
 --en --xml
```

<u>Notes:</u>
Invoking `java -jar` is possible only if the desired Java Runtime Environment (JRE) is the standard version.

If required, the Java Virtual Memory can be quickly adapted. -Xmx should be adjusted in "Out of Memory" and -Xss at "Stack Overflow" errors
(`java -Xmx1024m -Xss128m -jar`).

A command component that contains spaces needs to be enclosed in quotation marks.

KOST-Val can be invoked from any location. However this may require using absolute paths.

Building KOST-Val command:

```
Java -jar  cmdKOST-Val.jar-path  Val-type  file-path  [language]  [log type]
a--------  b-------------------  c-------  d--------  e---------  f---------
a = Java command
b = path and file cmdKOST-Val.jar
c = Type of validation:
    --format  (only format validation) or
    --sip     (SIP incl. format validation)
    --onlysip (only SIP validation)
d = path and full name of the file or folder being validated
e = language: --en or --de or --fr or --it (optional)
f = log type of the output (optional):
    --xml (standard log in xml format)
    --min (Log in xml format with only valid or invalid)
    --max (xml + verbose option if you want to keep the original reports)
```

3

The file has been validated as soon as "`Valid`" or "`Invalid`" is displayed in the command window. The folder has been validated as soon as the prompt (`C:\Program Files\KOST-CECO\KOST-Tools\KOST-Val>`) is displayed.

```
Format validation performed.
 -> From the 11542 files 553 (4.791%) are valid, 40 (0.347%) invalid and 10949 (94.862%) have not been accepted.
 -> C:\Users\clair\.kost-val_2x\logs\resources.kost-val.log.xml

C:\Program Files\KOST-CECO\KOST-Tools\KOST-Val>
```

Detailed results are available in the file kost-val.log.xml.

The overall result (valid/invalid file) is output as well. In addition, it is visible in the program's exit status in order for the validation to be embedded into an automated process chain. The exit status can take the following values:

- 0 everything is ok
- 1 incorrect program call
- 2 not valid

[3] To change the drive type, e.g., `c:`.

9 Copyright

Notice:	This product includes software developed by the Apache Software Foundation (apache.org/).

KOST-Val uses the following unmodified components of other manufacturers by embedding them directly into the source code:

Third party application / component	Version	License
3-Heights™ PDF/A Validator API pdf-tools.com	6.27.13.5	see Chapter 9.1
Apache Commons commons.apache.org/ - commons-compress - commons-logging - commons-io	 1.28.0 1.3.6 2.21.0	Apache License 2.0
Apache PDFBox pdfbox.apache.org/	3.0.7	Apache License 2.0
Apache Xerces2 xerces.apache.org/	2.12.2	Apache License 2.0
BadPeggy coderslagoon.com/	2.0	GPL v3 License
JACOB github.com/freemansoft/jacob-project	2.21	LGPL v2.1 License
Jdom jdom.org/	2.0.6.1	jdom License
Jhove openpreservation.org/tools/jhove/	1.34	LGPL v2.1 License
Spring Framework API spring.io/projects/spring-framework/	7.0.3	Apache License 2.0
veraPDF software.verapdf.org/dev/	1.31.17	GPL v3+ License
zip64 sourceforge.net/projects/zip64file/	1.02	GPL v2+ License

KOST-Val uses the following unmodified components of other manufacturers which are delivered with KOST-Val:

Third party application / component	Version	License
egov-validationclient-cli bit.admin.ch/	2.1.5	see Chapter 9.2
flac xiph.org/flac	1.5.0	BSD License
Jpylyzer openpreservation.org/tools/jpylyzer	2.2.1	LGPL v3.0 License
mkvalidator matroska.org/	0.6.0	BSD License
pngcheck libpng.org/pub/png/apps/pngcheck	3.0.3	GPL v2 License
GNU sed **gnu.org/software/sed**	4.4	GPL v3+ License
Xmllint xmllint.com/	20630	MIT License

Users of KOST-Val are requested to adhere to these components' terms of licence available in the folder KOST-Val\license.

9.1 3-Heights™ PDF/A Validator API License

For the use of the Restricted Version of the 3-Heights™ PDF/A Validator from PDF Tools KOST has made the following individual agreement on the General License Terms with PDF Tools:

> **2. Individuelle Vereinbarung**
>
> Dieses Vertragsverhältnis regelt die Client-Lizenz zwischen der PDF TOOLS als Lizenzgeber und der KOST als Lizenznehmer gemäss nachfolgenden Spezialbestimmungen:
>
> - PDF Tools AG erteilt für KOST eine kostenfreie OEM-Lizenz für das 3-Heights™ PDF/A Validator API als Zusatzfunktion ihrer eigenen Validator-Software (KOST-Val).
> - Die Lizenz schliesst den Gebrauch der Software (KOST-Val) durch Gedächtnisinstitutionen, bestehend aus Archiven oder Bibliotheken, deren Zulieferer und der KOST selbst, ein.
> - Der OEM-Lizenzschlüssel, welcher fest in KOST-Val eingebunden ist, darf nicht ausserhalb der Applikation (KOST-Val) verwendet werden.
> - Die Lizenz ist zeitlich unbegrenzt, jedoch bezüglich Durchsatz pro Installation begrenzt (72'000 Seiten pro Jahr).
> - Für die Verteilung der Software (KOST-Val) an den Anwender ist die KOST zuständig.
> - Der First Level Support der Anwender erfolgt durch KOST. Second Level Support Fälle leitet KOST an PDF Tools AG weiter.
> - Wenn der Anwender weitergehende Bedürfnisse hat, z.B. höherer Durchsatz, Integration in andere Applikationen etc. kauft er die Software (3-Heights™ PDF/A Validator API) direkt bei PDF Tools AG.
> - Die KOST darf weiterhin den Quellcode von KOST-Val Open Source publizieren und KOST-Val gratis und ohne Registrierung abgeben.

The following points are decisive for users:

- The license includes the use of the software (KOST-Val) by memory institutions consisting of archives or libraries, their suppliers and KOST itself.
- The OEM licence key, which is firmly integrated in KOST-Val, may not be used outside the application (KOST-Val).
- The license is unlimited in time, but limited in throughput per installation (72'000 pages per year).
- The first level support of the users is provided by KOST. Second Level Support cases are forwarded by KOST to PDF Tools AG.
- If the user has additional requirements, e.g. higher throughput, integration in other applications, etc., he or she can purchase the software (3-Heights™ PDF/A Validator API) directly from PDF Tools AG. The activation of this license is done with the “LicenseManager.exe”, which already exists in “KOST-Val\resources\3-Heights_PDF-Validator_dll”.

Users of KOST-Val are required to comply with this license agreement.

9.2 egov-validationclient-cli license

The discreet Validator only works with a cantonal license, which must be integrated into KOST-Val. If you wish to use the discreet Validator, please contact the KOST office (claire.roethlisberger@kost.admin.ch).

- If your institution is already registered, we will give you a small identification file which you must store in your configuration directory "USERHOME/.kost-val_2x/configuration".
- Cantonal archives that are not yet registered must ask their cantonal IT department whether they can send the user name and password to KOST.
  The office will provide you with a draft e-mail.
  The functionality can then be released for the State Archives and, if permitted, also for the canton's municipal archives.
- Municipal archives that are not yet registered should contact the KOST office directly so that the next steps can be defined.

10 Annex

10.1 Program structure

KOST-Val is structured according to the following requirements:

Functional requirements:

For every step the results (including information on inconsistencies and errors) are output and written into a validation log.

The overall result (valid/invalid file) is output as well. In addition, it is visible in the program's exit status in order for the validation to be embedded into an automated process chain. The exit status can take the following values:

- 0 everything is ok
- 1 incorrect program call
- 2 not valid

The validation steps are executed sequentially. Whenever possible the validation shall continue after an error has been detected in order to reduce the number of correction cycles.

Non-functional requirements:

External programs or java frameworks are used for particular tasks.

The application has a modular structure that allows for inserting additional validation modules without further ado.

The validation log and exit status permit an easy readout of a single validation result and allow the utilisation of the tool in a process chain.

The console output is limited on the validation module, the final results of either "valid" or "invalid" and the path to the file. All additional information is documented in the log file.

10.2 Functional Principle of Format Validation and Repair Function